

Learning the Right Boundaries: How Rejected-Answer Design Shapes Preference Tuning for Financial Causal Extraction

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Abstract

Financial causal extraction requires models to identify the exact text span that answers a causal question in a financial passage. This study examines whether Direct Preference Optimization (DPO), which trains a model to prefer correct answers over incorrect alternatives, can improve answer-boundary selection. Starting from a supervised fine-tuned Qwen3-4B-Instruct-2507 baseline (B1), we compare three DPO variants. Each DPO variant used 236 preference pairs. C1 used rejected answers taken from unrelated training questions. M1 used 223 incomplete and 13 overextended model-generated rejections. M2 retained 131 of these rejections—118 incomplete and 13 overextended—and replaced the remaining 105 incomplete rejections with 105 constructed overextensions, yielding a balanced dataset with 118 rejections of each type. The models were selected on a 196-example development set and evaluated on a frozen 391-example test set. C1 produced the same Exact Match as B1 (82.86%). M1 developed a tendency to lengthen answers and reduced Exact Match to 79.28% relative to B1 (nominal, unadjusted $p = 0.0385$). The M2 configuration no longer exhibited M1’s pronounced lengthening pattern and achieved 83.89% Exact Match, 94.44% token F1, and 92.99% Semantic Answer Similarity. M2 outperformed M1 (nominal, unadjusted $p = 0.0029$), although the primary comparison between M2 and B1 was not statistically significant ($p = 0.5413$). These results show that rejected-answer design strongly influences learned behavior. Because M2 changed both the balance and construction of the negatives, however, their individual effects cannot be isolated.

1. Introduction

Financial documents report not only what happened to a company but also why it happened. They may explain, for example, that revenue increased because of higher demand, expenses declined following a restructuring, or liquidity deteriorated because of increased borrowing. Automatically extracting these causal explanations could help analysts process large volumes of financial text more efficiently and support applications such as financial analysis, information retrieval, and summarization (Alqarni et al., 2026; Gautam et al., 2026).

The FinCausal shared task benchmarks this capability. In its question-answering formulation, a model receives a financial passage and a question about a cause or effect and must return the relevant span verbatim (Moreno Sandoval et al., 2025; Moreno-Sandoval et al., 2026). Success therefore requires both identifying the correct causal relationship and selecting its exact boundaries. An answer can convey the correct meaning but fail Exact Match because it omits necessary words or includes unnecessary surrounding text.

DPO offers a possible way to improve boundary selection by training a model to prefer the gold span over an incorrect alternative (Rafailov et al., 2023). However, what the model learns may depend on the types of errors represented by those alternatives. This study therefore asks: **How do the composition and construction of rejected answers in DPO training data affect answer-boundary behavior in financial causal extraction?**

Starting from a supervised fine-tuned Qwen3-4B-Instruct-2507 baseline (B1), we compare three DPO configurations using the same questions, preferred answers, pair counts, prompts, training settings, and random seed. C1 pairs each gold answer with a complete answer drawn from a different training question. M1 uses screened model-generated boundary errors that are heavily concentrated among incomplete spans. M2 retains 131 M1 rejections but replaces 105 incomplete rejections with constructed overextensions, producing equal numbers of incomplete and overextended negatives. Thus, M1 and M2 differ in both error-type balance and the source of part of their rejected answers.

The results show that generic negatives do not change Exact Match, while M1’s imbalanced preference data are associated with a systematic tendency to lengthen answers and lower performance. The balanced hybrid M2 configuration did not exhibit M1’s pronounced lengthening pattern and performed better than M1. However, the M2–M1 comparison was secondary, and its p -value was not adjusted for multiple comparisons. The primary comparison between M2 and B1 was not statistically significant. Because M2 changes both error-type balance and negative construction, the individual effects of these changes cannot be isolated.

2. Related work

2.1 Financial causal extraction

Early FinCausal editions included sentence level causality classification and token level extraction of cause and effect spans. Span extraction systems generally used transformer encoders, sometimes augmented with syntactic or graph-based

information. For example, Ziwei Xu et al. (2022) incorporated an external cause-effect knowledge graph into a BERT-based token classifier to help distinguish cause spans from effect spans. FinCausal 2025 reformulated the task as extractive question answering in English and Spanish: given a financial passage and a question about a cause or effect, systems returned an answer taken from the passage and were evaluated using Exact Match and Semantic Answer Similarity (Moreno Sandoval et al., 2025). FinCausal 2026 retained this formulation while adding more complex causal structures, reformulating some questions to require greater reasoning, and replacing Exact Match and Semantic Answer Similarity with LLM-as-a-judge evaluation for the official ranking (Moreno-Sandoval et al., 2026).

Recent system papers have investigated several approaches to this formulation. Trivedi et al. (2025) combined a fine-tuned extractive RoBERTa model with answer refinement using Gemma2-9B. Alqarni et al. (2026) evaluated five retrieval-augmented few-shot strategies with both base and fine-tuned GPT-4.1-mini models. Gautam et al. (2026) compared encoder-only, encoder-decoder, and decoder-only architectures using prompting, few-shot demonstrations, and supervised fine-tuning. In both 2026 studies, task-specific fine-tuning produced the largest overall gains, although exact extraction remained difficult in structurally complex passages.

2.2 Answer-boundary selection

Answer-boundary errors are not unique to financial text. Gangi Reddy et al. (2020) showed that machine-reading systems frequently return partially correct answers that are either incomplete or broader than the reference answer. They describe the desired answer as both minimal and sufficient and demonstrate that a second model trained to correct an initial predicted span can significantly improve performance.

Comparable problems appear in FinCausal systems. Alqarni et al. (2026) identify two main error types: slightly overextended or incomplete spans and causal-direction confusion when several cause-effect relationships occur in the same passage. Gautam et al. (2026) similarly report persistent errors in nested causal structures and passages containing multiple potential causes. They argue that limited annotation guidance compounds the ambiguity and that prompt optimization alone cannot fully resolve it. Thus, strong semantic performance did not consistently translate into exact span extraction.

2.3 Learning from rejected answers

Prior question-answering research demonstrates that negative-example construction can affect what a model learns. Lee and Lee (2023) used supervised contrastive learning—not DPO—to distinguish the correct extractive span from competing candidate answers generated by a pretrained QA model. DPO provides a different mechanism for generative models, increasing the relative likelihood of a preferred response over a rejected response against a frozen reference model (Rafailov et al., 2023).

Preference pairs have been constructed in several ways in other domains. For machine translation, Haoran Xu et al. (2024) form triplets containing a human reference, a GPT-4 translation, and an ALMA translation. Automatic quality estimators identify the highest-scoring translation as preferred and the lowest-scoring translation as rejected. Their results show that even relatively high-quality but imperfect translations can provide a useful preference signal. Tunstall et al. (2023) use a different strategy in Zephyr, training on responses generated by multiple language models and ranked through AI feedback. These studies demonstrate the usefulness of automatically constructed preference pairs, but they do not organize rejected responses according to controlled, task-specific error categories.

Research has also established that DPO can amplify length-related signals. Park et al. (2024) show that even a modest preference for longer responses in dialogue and summarization datasets can lead DPO models to produce substantially longer outputs. They introduce length regularization to limit this behavior. Our study examines a related but narrower problem in exact-span extraction: whether repeatedly preferring a gold span over an incomplete span induces a directional bias toward longer answers, and whether including an opposing set of overextended rejections prevents that bias.

Gautam et al. (2026) identify preference-alignment methods such as DPO as a promising direction for learning the implicit span conventions in FinCausal. To our knowledge, however, no prior FinCausal system has applied DPO or compared different rejected-answer designs. We address this gap by comparing unrelated rejected answers, screened model-generated boundary errors dominated by incomplete spans, and a balanced hybrid dataset containing both retained model-generated errors and constructed overextensions. The configurations use the same questions, preferred answers, pair counts, and training settings, but M1 and M2 do not use the same negative-construction or validation procedure. Their comparison therefore shows how these two preference-data designs affect extraction behavior but cannot completely separate the effect of error-type balance from the effect of constructing overextended negatives.

3 Data

3.1 FinCausal dataset and splits

We use the 2,000 labeled English examples released for FinCausal 2026 (Moreno-Sandoval et al., 2026). Each example contains a financial passage, a question asking about a cause or effect, and a gold answer copied verbatim from the passage. Only the English data are used in this study.

Because the same passage can support multiple questions, examples were split by passage rather than independently by row. This produced initial training, development, and test sets of 1,400, 200, and 400 examples. We then audited passages across splits using normalized text matching and character-level similarity. Manual review identified 13 overlapping or near-duplicate evaluation examples. We removed the later-occurring examples—four from development and nine from test—while leaving the training set unchanged.

Split	Initial size	Removed	Final size
Training	1,400	0	1,400
Development	200	4	196
Test	400	9	391

3.2 Preference datasets

Each DPO example contains the original passage and question, the gold answer as the preferred response, and an incorrect answer as the rejected response. All preference pairs were constructed exclusively from the training split. The 236 examples were not selected in advance; they represent the unique training questions for which candidate generation and screening produced at least one acceptable B1 boundary error. C1, M1, and M2 use the same 236 questions and preferred answers; only the rejected answers differ.

M1 was constructed from boundary errors generated by the supervised B1 model. Candidate answers were screened to identify incomplete spans, which omitted part of the gold answer, and overextended spans, which included unnecessary surrounding text. A blind audit confirmed all 50 sampled screened candidates as valid boundary errors. After retaining one candidate per question, M1 contained 223 incomplete and 13 overextended rejected answers.

C1 provides a generic-negative control. For each M1 question, the rejected response was replaced with a complete gold answer from a different training example. Donor answers were matched to the corresponding M1 rejection by length and, where possible, causal role. A separate audit confirmed all 50 sampled C1 matches as valid rejected answers.

M2 was constructed after development-set analysis revealed M1’s tendency to lengthen answers. It retained 118 incomplete and the 13 original overextended M1 rejections. The remaining 105 incomplete rejections were replaced with overextended spans constructed by adding adjacent passage text to the gold answer at natural clause or sentence boundaries. This produced equal numbers of the two opposing boundary errors without changing the questions, preferred answers, or number of pairs.

Dataset Rejected-answer construction		Incomplete	Overextended	Total
C1	Complete answer from another training question	N/A	N/A	236
M1	Screened B1 boundary errors	223	13	236
M2	Balanced hybrid: retained model errors and constructed overextensions	118	118	236

4 Methodology

4.1 Model training

We use Qwen3-4B-Instruct-2507 as the base model. The baseline model, B1, was supervised fine-tuned on the 1,400 training examples using QLoRA (Dettrmers et al., 2023). B1 was trained for three epochs with a learning rate of 1×10^{-4} , an effective batch size of 8, a maximum sequence length of 640 tokens, and random seed 42. QLoRA used 4-bit NF4 quantization with LoRA rank 16, $\alpha = 32$, and dropout 0.05. C1, M1, and M2 were trained independently from the B1 checkpoint using DPO, with B1 also serving as the frozen reference model. All three DPO configurations used the same questions, preferred answers, prompt format, training settings, and random seed; only the rejected answers differed, as described in Section 3.2.

4.2 Experimental procedure

B1, C1, and M1 were first compared on the development set. The directional lengthening observed in M1 motivated the balanced M2 configuration. After the M2 design was fixed, the models were evaluated on the frozen test set. All models used the same prompt and greedy decoding at inference.

5 Evaluation

5.1 Evaluation metrics

Strict Exact Match (EM) is the primary metric and requires the prediction to match the gold answer exactly after removing surrounding whitespace. We additionally report token F1 and Semantic Answer Similarity (SAS), computed using cross-encoder/stsb-roberta-large (Risch et al., 2021).

For boundary analysis, predictions remaining incorrect after conservative normalization were labeled *too short* when the prediction was contained within the gold answer and *too long* when the gold answer was contained within the prediction. Semantic correctness was evaluated using a hybrid procedure. Predictions matching the gold answer after normalization were accepted automatically. The remaining predictions were evaluated by a blinded GPT-5.6-sol judge, which received anonymized predictions without model identities and assigned an adequacy score from 1 to 5; scores of 4 or 5 were accepted. Hybrid semantic acceptance is the number of automatically accepted and judge-accepted predictions divided by all 391 test examples.

5.2 Statistical testing

Because all systems answered the same 391 test questions, comparisons were paired. M2 versus B1 was the predeclared primary comparison. All comparisons were evaluated on strict EM using two-sided exact McNemar tests, and 95% confidence intervals for paired EM differences were estimated using 10,000 paired bootstrap resamples. C1 versus B1, M1 versus B1, and M2 versus M1 were secondary comparisons, while M2 versus C1 was exploratory. The p -values for all non-primary comparisons are nominal and unadjusted for multiple testing.

6 Results

6.1 Development results

On the development set, B1 achieved 83.67% EM, C1 83.16%, and M1 79.59%. After balancing the rejected-answer types, M2 achieved the highest development EM at 84.18% and was selected for the frozen test evaluation.

6.2 Test results

Table 1 reports performance on the 391-example test set. M2 obtained the highest score across all four metrics, while M1 performed worst.

Model	Exact Match	Token F1	SAS	Hybrid semantic acceptance
B1	82.86	93.26	92.12	91.82
C1	82.86	93.37	92.39	91.05
M1	79.28	92.19	91.48	90.79
M2	83.89	94.44	92.99	93.35

Table 1: Test performance on a 0–100 scale.

Table 2 presents the principal paired EM comparisons. C1 produced no aggregate change relative to B1. M1 performed 3.58 percentage points below B1, whereas M2 outperformed M1 by 4.60 points. The predeclared comparison between M2 and B1 showed a smaller improvement of 1.02 points that was not statistically significant. M2 exceeded C1 by 1.02 percentage points in Exact Match, but the difference was not statistically significant (95% bootstrap CI $[-1.53, 3.58]$; McNemar $p = 0.5716$).

Comparison	EM difference (pp)	95% paired CI	McNemar (p)
C1 – B1	0.00	$[-2.05, 2.05]$	1.0000
M1 – B1	–3.58	$[-6.65, -0.51]$	0.0385
M2 – M1	+4.60	$[1.79, 7.67]$	0.0029
M2 – B1	+1.02	$[-1.53, 3.58]$	0.5413
M2 – C1	+1.02	$[-1.53, 3.58]$	0.5716
M1 – C1	–3.58	$[-6.91, 0.00]$	0.0649

Table 2: Paired comparisons in strict Exact Match. M2 versus B1 was the predeclared primary comparison. The secondary and exploratory comparison p -values are nominal and unadjusted.

7 Error Analysis and Discussion

7.1 Span patterns and negative mining

Table 3 groups predictions according to their automatic string relationship with the gold answer. Shorter predictions are contained within the gold answer, while longer predictions contain the gold answer. Other includes predictions with neither relationship, such as partial overlaps, non-verbatim answers, and different spans. These categories describe string overlap with the annotated answer and do not by themselves determine semantic correctness.

	Model	Exact	Non-exact	total	Shorter	Longer	Other
B1	324	67			25	34	8
C1	324	67			31	30	6
M1	310	81			9	63	9
M2	328	63			22	35	6

Table 3: Automatic span-relation categories on the 391-example test set.

Before constructing M1, an audit of B1’s development errors found that shorter and longer boundary errors were approximately balanced. However, generating and screening candidate answers for preference training produced 236 usable boundary negatives, of which 223 were incomplete and only 13 were overextended. We retained this distribution because M1 was intended to use authentic model-generated hard negatives. The resulting imbalance therefore did not simply reflect B1’s observed development errors; it emerged from the negative-mining pipeline. The present analysis cannot determine whether the skew arose primarily during candidate generation or screening.

M1’s behavior closely followed this one-sided training signal. Compared with B1, shorter errors fell from 25 to 9, but longer errors increased from 34 to 63, reducing EM from 82.86% to 79.28%. M1 learned to avoid incomplete answers but generalized this preference into a broader tendency to produce longer spans. C1 provided no measurable improvement over B1 in exact boundary selection. Although B1 and C1 had identical Exact Match, their predictions were not identical: C1 corrected eight B1 errors and introduced eight new errors, while 59 of the 67 non-exact cases were shared. This substantial overlap indicates that generic negatives did not systematically resolve B1’s existing boundary errors

The hybrid M2 configuration reduced longer errors from 63 in M1 to 35 and produced the highest EM. However, M2 achieved balance by replacing 105 naturally generated incomplete errors with constructed overextensions. The M1–M2 comparison therefore cannot establish that balance alone produced the change, because negative provenance and construction method changed simultaneously. The combined evidence nevertheless shows that the design of preference data matters: the one-sided mined dataset was associated with a strong lengthening bias, whereas the balanced hybrid dataset was not.

7.2 Prediction transitions and qualitative errors

Aggregate EM conceals which individual predictions changed. M2 corrected 14 B1 errors but made 10 previously exact B1 answers non-exact, resulting in a net gain of four answers.

B1 → M2 transition	Cases Span changes	
Non-exact → exact	14	8 too-short B1 spans were expanded; 4 too-long B1 spans were trimmed; 1 partial-overlap and 1 different-span prediction were corrected
Exact → non-exact	10	5 M2 predictions became too short; 5 became too long
Net change	+4	4 additional exact answers

Table 4: Strict Exact-Match transitions between B1 and M2.

Manual review found that seven of the ten regressions remained semantically acceptable despite failing EM. The other three were substantive errors: two incomplete answers and one overextension that added an unrelated contrasting clause. More broadly, the hybrid semantic evaluation accepted 37 of M2’s 63 strict Exact-Match failures, either through normalized matching or GPT evaluation.

The following examples illustrate what M2 corrected, what it damaged, and what M2 did not resolve.

M2 corrected B1’s selection of the wrong relationship.

Passage: “Macfarlane Group uses total turnover (£000) in the reporting period to calculate the intensity ratio, as this allows emissions to be monitored over time taking into account changes in the size of the Group. This factor was chosen because it provides the greatest degree of accuracy.”

Question: *What led them to choose total turnover as the factor?*

Gold and M2: “it provides the greatest degree of accuracy”

B1: “this allows emissions to be monitored over time taking into account changes in the size of the Group”

B1 selected the effect of using total turnover rather than the reason it was chosen. M2 distinguished between the two causal relationships and returned the correct span.

M2 overextended an exact B1 answer.

Passage: “He was unable to attend one meeting (due to illness) but received the papers.”

Question: *What was the outcome of Richard Baker’s illness?*

Gold and B1: “He was unable to attend one meeting”

M2: “He was unable to attend one meeting (due to illness) but received the papers”

M2 included the correct outcome but added a contrasting detail that was not caused by the illness. This was a genuine B1-to-M2 regression, not merely an alternative acceptable boundary.

Both models reversed the causal direction.

Passage: “RWAs reduced by £27 billion (12%), ending the year at £201 billion (from £228 billion in 2016). This reduction was driven by NatWest Markets, where RWAs fell by £16.8 billion due to the continued run-off of the legacy business and mitigation activities.”

Question: *What did the reduction in RWAs by £27 billion lead to?*

Gold: “ending the year at £201 billion (from £228 billion in 2016)”

B1: “NatWest Markets, where RWAs fell by £16.8 billion”

M2: “NatWest Markets, where RWAs fell by £16.8 billion due to the continued run-off of the legacy business and mitigation activities”

The question asks for the effect of the reduction, but both models returned its source or causes. Other reviewed cases similarly confused a cause with a stated purpose or included concessive information that opposed the identified cause. These errors show that the balanced hybrid M2 configuration did not exhibit M1’s pronounced length bias but did not systematically improve deeper causal reasoning.

7.3 Limitations and future work

The experiment uses only 236 preference pairs, one base model, and English data from a single financial causal-extraction dataset. M2 was designed after observing M1’s development behavior, although the test set remained frozen until the M2 design was fixed. In addition, 105 of M2’s overextended negatives were artificially constructed by appending adjacent passage text to the gold answer, rather than obtained from naturally generated model errors. These negatives may therefore differ from real model overextensions in their plausibility and difficulty. Finally, M2’s improvement over B1 was not statistically significant.

The main limitation is that M2 differed from M1 in more than error-type balance. M1’s rejected answers were generated by B1, screened as boundary errors, and heavily skewed toward incomplete answers. To create M2, 105 incomplete M1 rejections were replaced with artificially constructed overextensions. These constructed negatives received a structural audit but not the same sampled human semantic validation used for M1, and they may differ from natural model errors in plausibility and difficulty. Therefore, the experiment cannot show whether M2’s behavior was due to the balanced error distribution, the use of constructed negatives, or both. A more controlled study would compare balanced and imbalanced datasets created and validated using the same procedure.

Because the experiment was designed to isolate answer-boundary behavior, the preference data did not systematically target purpose–cause confusion, causal-direction reversals, concessive contamination, or selection among multiple causal relationships. Future work could construct controlled preference pairs for these errors and evaluate them on a new held-out dataset. The present evidence supports a narrower conclusion: the balanced hybrid intervention did not exhibit the directional length bias associated with M1’s imbalanced preference data, but it did not resolve all errors in financial causal extraction.

8 Conclusion

This study examined how rejected-answer design affects DPO training for financial causal extraction. Unrelated negatives in C1 produced no improvement over the supervised baseline. The naturally mined M1 negatives were heavily concentrated among incomplete answers, and M1 developed a corresponding lengthening bias: shorter errors declined, but longer errors increased substantially, reducing Exact Match. The balanced hybrid M2 configuration did not exhibit this bias and produced the highest overall performance, although its improvement over B1 was not statistically significant.

The strongest finding is that naturally mined hard negatives are not automatically suitable preference data. Their distribution should be examined before training because a one-sided signal can produce a predictable but undesirable behavioral shift. However, because M2 changed both error-type balance and negative construction, this experiment does not establish that balance alone caused its improved behavior. Future work should compare balanced and imbalanced datasets created

through a common construction and validation procedure and should include controlled negatives targeting causal direction, purpose, concessions, and competing relationships.

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